Pradyumna Yerabati Venkata

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Education

Texas A&M University, College Station, TX — GPA: 4.00/4.00
Master of Science in Computer Science — *Jan 2026 – Dec 2027*

National Institute of Technology, Warangal, India
Bachelor of Technology in Electrical and Electronics Engineering — *Jun 2019 – May 2023*

Professional Experience

Texas A&M University — Mar 2026 – Present
Student Research Assistant — Advisor: Prof. Shuang Zhang — *College Station, TX*

- Designed and developed data integration pipelines for large multi-source hydrological datasets, handling missing values, temporal alignment, and spatial aggregation across thousands of monitoring stations.
- Designed Graph Neural Network architectures for spatio-temporal prediction of river alkalinity, outperforming traditional tabular baselines across multiple evaluation metrics.
- Built LSTM temporal baselines and conducted GPU-accelerated hyperparameter tuning on Texas A&M HPRC clusters to optimize large-scale deep learning experiments.
- **Tech Stack:** Python, PyTorch, Graph Neural Networks, LSTM, NumPy, Pandas, Scikit-learn.

HCL Technologies — Jul 2023 – Jan 2026
Technical Lead — *Bangalore, India*

- Optimized BigFix IVR backend systems handling 100,000+ clients, reducing processing time from 3 days to 3 hours — a 90% efficiency gain.
- Led code reviews and pull request workflows on GitHub, enforcing branch protection and Git-based development practices within an agile delivery model.
- Engineered Flask-based mock server architectures and integrated them into Jenkins CI/CD pipelines, cutting backend validation overhead by 60% and accelerating release iteration.
- Implemented a Redis-based API rate limiter to guarantee system availability and prevent overload under peak traffic.
- Built an AI-powered PDF translation workflow using LangChain and LLaMA LLMs, applying prompt engineering, output validation, and human-in-the-loop review to improve multilingual extraction accuracy by 70%.
- **Tech Stack:** Java, Python, Flask, JavaScript, Node.js, REST APIs, MySQL, Redis, MongoDB, SQLite, Docker, Kubernetes, GCP, Jenkins, Prometheus, Grafana, OpenAPI, Swagger.

Projects

WisdomLinked | *Node.js, Express.js, MongoDB, React.js, Docker* — wisdomlinked.com

- Contributing to a production full-stack mentorship platform connecting students with domain experts, supporting session booking, real-time messaging, video conferencing (Jitsi), and Stripe/PayPal payments.
- Built and maintain RESTful backend APIs with Node.js, Express.js, and MongoDB, covering JWT auth, session management, OTP-based 2FA, and OAuth (Google, Facebook, Twitter).
- Integrated Digital Ocean Spaces for scalable cloud object storage, implementing secure upload and retrieval pipelines via the AWS S3-compatible SDK.
- Containerized deployments with Docker and Docker Compose behind an Nginx reverse proxy, configuring multi-environment setups across local and production.

Intelligent Whiteboard Agent | *React.js, Node.js, OpenAI API, Excalidraw, WebSockets*

- Built an AI-assisted development tool: an Excalidraw whiteboard with an embedded chat agent that generates architecture diagrams and visual workflows from natural language prompts.
- Implemented real-time WebSocket communication and structured prompt workflows to convert user instructions into precise diagram components.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL
Frontend: React.js, Next.js, HTML5, CSS3, Redux/Tailwind CSS
Backend: Spring Boot, Node.js, Express.js, REST APIs, GraphQL, gRPC, Microservices
Databases: PostgreSQL, MySQL, MongoDB
Event-Driven & Messaging: Kafka, Redis
Cloud & DevOps: Docker, Kubernetes, GitHub Actions, Jenkins, CI/CD, AWS, GCP
AI-Assisted Development: GitHub Copilot, Cursor, LangChain, LLM APIs, Prompt Engineering
Testing & Tools: JUnit, Jest, Cypress, Postman, Swagger, Git, Grafana, Prometheus
System Design: Distributed Systems, API Design, Scalability, Caching, Load Balancing, Observability, Authentication/Authorization

Achievements

Published engineering optimization case study *From 3 Days to 3 Hours* in the HCL Central Engineering Newsletter.
Solved 1000+ algorithmic problems on LeetCode and GeeksforGeeks; achieved Gold Badge and 5-Star SQL Rating on HackerRank.
Received monetary performance awards at HCL Technologies for exceptional technical contributions.